Interim Report

Team Project Module · National College of Ireland · Individual Project

Project Name: PrepCast
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Module / Cohort: Team Project Module (Individual)
Date: 3 July 2026

1. Project Overview

PrepCast is a Machine Learning web application for managers of food service and fulfilment centres.

The system forecasts next week's meal demand, and converts it into a recommended prep plan, to help managers plan production, meet demand, and reduce waste.

2. Deliverables & Progress

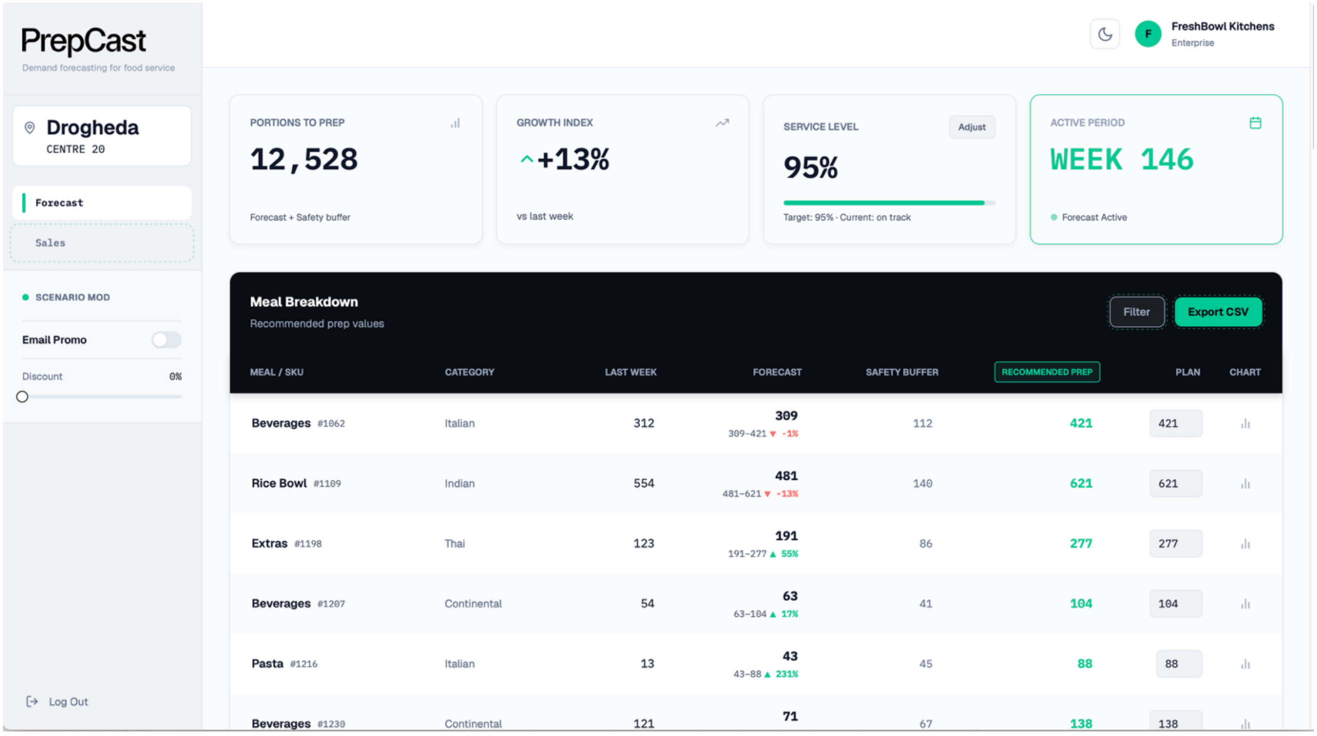
Deliverable	Status	Notes (what remains; will you reach it?)
Project proposal	Done	Submitted, Dataset sourced, chosen from several options. Domain finalised.
Technical requirements SRS and initial project work	Done	SRS Submitted. Similar domain reports researched to gain insight for food prep forecasting, Several key findings extracted for use in project. e.g time-based training/testing, lag and rolling features, and safety buffer formula. Datasets columns and features explored and target value defined. Jupyter Notebook created with findings, charts and discoveries.
Sprint 1	Done	Schema ERD designed in Paradigm from a 3NF normalisation based off the dataset columns. Feature engineering used to create features: lag (previous weeks), rolling (average of previous weeks), and discount. Time based train/validation/test split on regression models (Linear regression, RandomForest, HistGradientBoosting), and scored against baselines of last week and 3 week moving average. Multicollinearity tested. Lag and rolling ranked highly correlated but as features overlapped, this was expected. Created SQLite DB from seed.py and schema.sql Features.py from notebook for feature engineering Investigated hyperparameters in RandomForest and GradientBoosting to find the optimal settings, which showed RandomForest highest scoring Saved highest R^2 scoring trained model (RandomForest) as model.joblib using train.py RMSE: 101.17356928209159 MAE: 60.75565417194504 R^2: 0.8662661849082801

Deliverable	Status	Notes (what remains; will you reach it?)
Sprint 2	Done	Created a safety buffer from standard deviation statistics formula and added to forecasting figures to create recommended prep results for each meal per location Issues with result begin flat, so a limit was placed on the number of weeks to 10 when calculating the standard deviation in formula. Create Flask API endpoints from the SQLite database to serve a frontend UI (app.py), /api/centres, /api/forecast, /api/recommend Promotion and discount factors added to predictions. Build of boilerplate frontend html/tailwind/js to receive data to display, for proof of concept. Designed a frontend UI in Figma Implemented design Added charts Deployed v1 to PythonAnywhere

3. Progress Against the MVP

On track and in a good place for functioning MVP. MUST requirements passed so far: FR-001 Suitable Model trained. FR-002 Promotions and discount features affect prediction. FR-003 Location login on frontend shows only that locations meals. FR-004 per-meal forecast and recommended prep listed with safety buffer.FR-005 Manager can adjust final prep plan amounts and save to the database.

4. Evidence / Current State



Repository link: https://github.com/davehaughton/prepcast-project
Trello link: https://trello.com/b/20SSVLXr/prepcast

Dataset: https://www.analyticsvidhya.com/datahack/contest/genpact-machine-learning-hackathon-1/
Model Evaluation: RMSE: 101.173, MAE: 60.756, R^2: 0.866
Deploy link: https://davehaughton.pythonanywhere.com

5. Challenges & Risks

Lag and rolling engineered features created NAN values, so rows with missing values were dropped as the model can't train on missing values.

When calculating lag weeks and rolling averages, only past weeks needed to be included. The current week needed to be excluded and a time-based train/test split was used, instead of random splitting.

When splitting Training/Validation/Testing, random splitting wasn't a suitable method as it was prone to leakage, so a time based splitting was used instead. Training: weeks 1-125, validation and hyperparameter testing weeks 126-135, and test on the remaining 136-145.

One of the original csv files called test.csv from the hackathon dataset had no target value, so it was omitted from training.

The safety buffer initially used all the weeks history when calculating the standard deviation. This gave a poor result and didn't match the recent weeks. Fixed this by limiting the weeks it used in calculations to 10 most recent weeks.

FR-006 Manager to save new data to database. There is a potential risk of data integrity and model re-training issues if saved data is very skewed. To mitigate this potential issue, the inputs will be validated, and empty inputs removed before retraining.

6. Plan to Completion

Week(s)	Planned Work
Wk 8	*SHOULD: Add saving of actual sales of current week. COULD: Add ability of the system to retrain the model on the actual sales so the model is up-to-date. SHOULD: show the current sales versus the prediction, to show how accurate the system forecast.*
Wk 9	*Iterate and polish the frontend UI and write the report*
Wk 10	*Finish the final report*